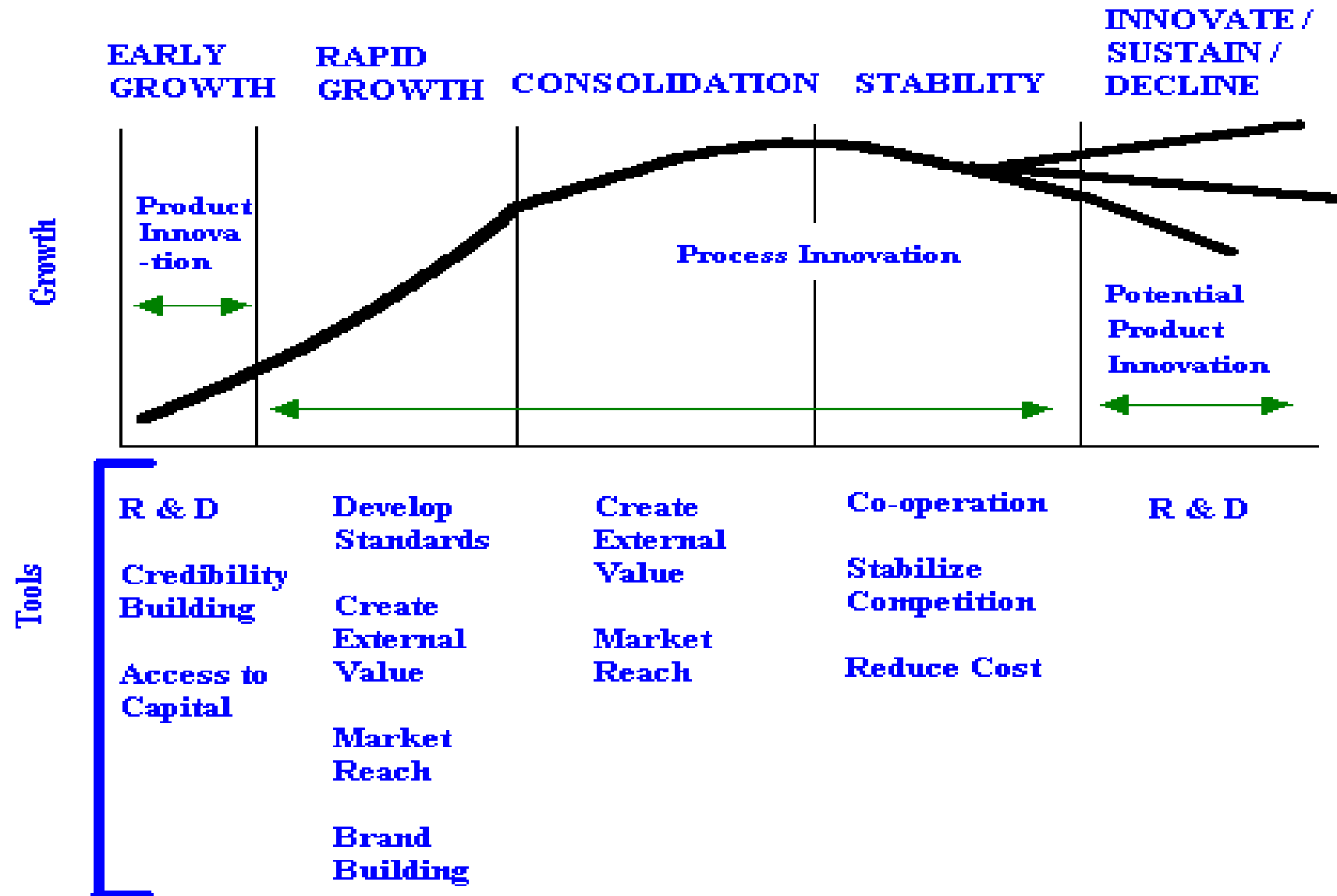


# Industry Analysis

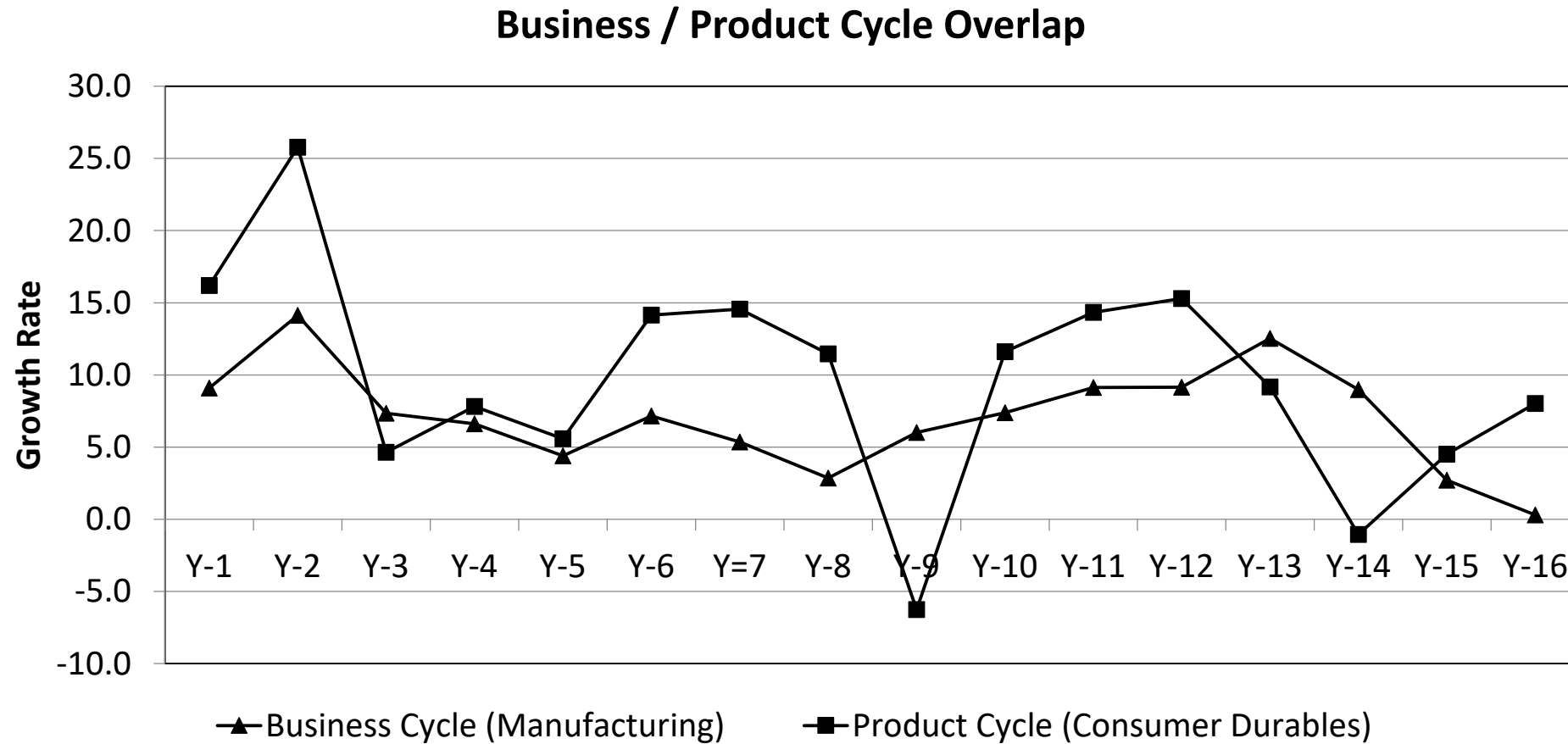
# Industry Definition

- **Economic sector** - manufacturing, distribution (wholesale or retail), services, etc.
- **Products or services** - a list of what is offered by the industry
- **Geographic scope** - local, regional, national, or international (This is for the overall industry and is determined by the markets in which products or services are offered by most competitors, not by the size or scope of individual companies.)

## BUSINESS LIFE CYCLE / PRODUCT LIFE CYCLE OVERLAP



# Growth Cycle



## Prediction of Market Industry Life Cycle Stage

|                                    | Introduction                                                                                                             | Growth                                                                                                                            | Maturity                                                                                                                                                                | Decline                                                      |
|------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| <b>Buyers and buyer behaviour</b>  | High income group purchases. Buyer inertia. Buyers must be convinced to try the product.                                 | Widening buyer group. Consumer accepts uneven quality.                                                                            | Mass market. Saturation in demand. Repeat buying. Choice among brands.                                                                                                  | Customers are sophisticated buyers of the product.           |
| <b>Products and product change</b> | Poor quality. Product design and development stage. Different product variations. No standards. Frequent design changes. | Products have technical and performance differentiation. Reliability and guarantee for products. Competitive product improvement. | Superior quality. Less product differentiation due to standardization. Less rapid product changes - more minor annual model changes. Trade channels become significant. | Little product differentiation. Doubtful product acceptance. |

## Prediction of Market Industry Life Cycle Stage

|                                       | Introduction                                                                                                                     | Growth                                                                                                      | Maturity                                                                                                                                           | Decline                                                         |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <b>Marketing</b>                      | High advertising / sales. Creaming price strategy.                                                                               | High advertising, but lower percent of sales than introductory phase. Distribution costs high for products. | Market segmentation. Efforts to extend life cycle. Broad product line. Service and deals prominent. Packaging important. Advertising costs lower.  | Low advertising and marketing costs.                            |
| <b>Manufacturing and Distribution</b> | Unutilized production capacity, Short production runs. High skilled labour content. High production costs. Specialized channels. | Full production capacity. Shift towards mass production. Focus on distribution. Mass channels.              | Optimum capacity. Stability of manufacturing process. Lower labour skills. Lower production runs. High physical distribution costs. Mass channels. | Substantial over-capacity. Mass production. Specialty channels. |
| <b>R&amp;D</b>                        | Changing production techniques. High R& D Cost.                                                                                  | Moderate R& D Cost.                                                                                         | Low R& D Cost.                                                                                                                                     | Nil R& D Cost.                                                  |
| <b>Foreign Trade</b>                  | Few exports.                                                                                                                     | Significant exports. Few imports.                                                                           | Falling exports. Significant imports.                                                                                                              | No exports. Significant imports.                                |

# Prediction of Market Industry Life Cycle Stage

|                            | <b>Introduction</b>                             | <b>Growth</b>                                                                            | <b>Maturity</b>                                                                                                      | <b>Decline</b>                                      |
|----------------------------|-------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>Business strategy</b>   | Low market share.<br>Holding large asset value. | Change in price or quality image.<br>Rapid asset acquisition.                            | Competitive pricing & cost reduction is the focus. No change in price, image or quality. No change in asset holding. | Cost control prominent. Sale or disposal of assets. |
| <b>Competition</b>         | Fewer companies.                                | Many competitors.                                                                        | Price competition.<br>Cyclical in sales.                                                                             | Fewer competitions.                                 |
| <b>Risk</b>                | High risk.                                      | High Risk.                                                                               | Moderate Risk.                                                                                                       | High Risk.                                          |
| <b>Margins and Profits</b> | High prices and margins. Low profits.           | High profits, highest margins, high prices but lower than introductory phase. High P/Es. | Falling prices, lower profits, lower margins, lower dealer margins. Stability of P/Es.                               | Low profits and margins, falling prices.            |

# Life Cycle Analysis: Variables

| Variables         | Measure                                     | Start up | Growth   | Maturity | Decline    |
|-------------------|---------------------------------------------|----------|----------|----------|------------|
| Buyers behaviour  | Sales growth                                | Low      | High     | Moderate | Low        |
| Production Return | Material cost / total cost                  | Low      | Moderate | High     | Negligible |
|                   | Operating margin                            | Low      | Moderate | High     | Negligible |
| Marketing         | Advertisement cost/Sales value              | High     | Moderate | Low      | Low sales  |
| Operating cost    | Labor cost/sales                            | High     | Moderate | Low      | Low sales  |
|                   | Distribution cost growth                    | High     | Moderate | Low      | Low        |
| R & D             | R&D Cost/total cost                         | High     | Moderate | Low      | Low        |
| Foreign Trade     | Foreign sales/total sales                   | Low      | High     | Moderate | Low        |
|                   | Imports/total costs                         | Low      | Low      | High     | Moderate   |
|                   | Cash flows / Operating profit               | Low      | High     | Moderate | Low        |
| Business strategy | Manufacturing cost/total cost               | High     | High     | Moderate | Low        |
|                   | Total assets growth rate                    | High     | Moderate | Moderate | Low        |
| Competition       | HHI index (sales)                           | Low      | High     | Moderate | Low        |
| Risk              | Coefficient of variation of sales (3 years) | High     | High     | Moderate | High       |
| Margins & Profit  | Net profit/sales                            | Low      | High     | Moderate | Low        |
|                   | Growth in profits                           | Low      | High     | Moderate | Low        |



# Life Cycle Analysis: Cyber Security

|                  | Sales growth | Operating margin | Advertisement /sales | Manufacturing cost % | R& D cost | Cash flows/op | Asset growth rate | HHI    | Cov sales | Growth in profits |
|------------------|--------------|------------------|----------------------|----------------------|-----------|---------------|-------------------|--------|-----------|-------------------|
| L&T Technology   | 6.76%        | 18.68%           | 12%                  | 14%                  | 0.37%     | 104%          | 11.20%            | 0.11   | 7.73%     | -0.96%            |
| Tata Technolog.  | 3.73%        | 17.32%           | 5%                   | 13%                  | 0.53%     | 94%           | 26.03%            | 0.03   | 4.02%     | -10.24%           |
| Inventus Knowl   | 32.55%       | 30.49%           | 11%                  | 5%                   | 0.39%     | 65%           | 11.90%            | 0.01   | 27.12%    | 39.69%            |
| Netweb Technol.  | 73.68%       | 13.73%           | 4%                   | 0%                   | 0.48%     | 50%           | 91.95%            | 0.00   | 55.53%    | 64.64%            |
| Affle 3i         | 21.24%       | 21.12%           | 27%                  | 41%                  | 3.59%     | 95%           | 16.22%            | 0.01   | 19.05%    | 23.77%            |
| Sagility         | 23.01%       | 23.63%           | 8%                   | 6%                   | 1.32%     | 99%           | 8.74%             | 0.05   | 21.26%    | 101.42%           |
| Amagi Media Labs | 30.89%       | -12.04%          | 49%                  | 1%                   | 0.00%     | 3%            | 34.12%            | 0.00   | 26.55%    | -276.18%          |
| TechD Cybersec.  | 86.19%       | 36.62%           | 17%                  | 16%                  | 40.00%    | 21%           | 207.32%           | 0.00   | 57.56%    | 116.02%           |
| Cyber security   | 34.76%       | 18.70%           | 16.57%               | 11.83%               | 5.83%     | 66.29%        | 50.94%            | 20.47% | 27.35%    | 7.27%             |
| IT               | 25.45%       | 20.45%           | 10.45%               | 10.35%               | 3.73%     | 78.84%        | 32.76%            |        | 20.51%    | 15.76%            |
| Stage            | High         | Low              | High                 | Moderate             | High      | Low           | High              | Low    | High      | Low               |

# Life Cycle Analysis: Cyber Security

| Variables         | Stage                                       | Introduction | Growth   | Maturity | Decline    |
|-------------------|---------------------------------------------|--------------|----------|----------|------------|
| Buyers behaviour  | Sales growth                                | Low          | High     | Moderate | Low        |
| Production Return | Operating margin                            | Low          | Moderate | High     | Negligible |
| Marketing         | Advertisement cost/Sales value              | High         | Moderate | Low      | Low sales  |
| Operating cost    | Manufacturing cost/sales                    | High         | Moderate | Low      | Low sales  |
| R & D             | R&D Cost/total cost                         | High         | Moderate | Low      | Low        |
| Cash flows        | Cash flows / Operating profit               | Low          | High     | Moderate | Low        |
| Business strategy | Total assets growth rate                    | High         | High     | Moderate | Low        |
| Competition       | HHI index (sales)                           | Low          | High     | Moderate | Low        |
| Risk              | Coefficient of variation of sales (3 years) | High         | High     | Moderate | High       |
| Margins & Profit  | Growth in profits                           | Low          | High     | Moderate | Low        |

# SWOT Analysis

|                      |                   |                      |                |
|----------------------|-------------------|----------------------|----------------|
| External Environment | Threats (T)       | Confront             | Avoid          |
|                      | Opportunities (O) | Exploit              | Search         |
|                      |                   | Strengths (S)        | Weaknesses (W) |
|                      |                   | Internal Environment |                |

# Impact of Internal Strategies

## **Weakness**

- Industry may not be in the wish list of investors
- Lends may identify a industry with low growth prospects
- Poor financial resources

## **Strength**

- Firms estimate cash flows for long duration
- Firms incorporate strategies to match cash inflows and outflows
- Resource Stability

# Internal Factors

|                  | Weakness                            | Strength                                                     | Overall Impact<br>(Weakness/Strength)         |
|------------------|-------------------------------------|--------------------------------------------------------------|-----------------------------------------------|
| <b>Ideal</b>     |                                     |                                                              |                                               |
| Strategy         | No impact (low risk)<br>Score: 0.5  | Impact (Low risk)<br>Score (0.5)                             | Stable: Score [1] (0.5 / 0.5)                 |
| Example          | Poor financial resources            | Resource stability                                           |                                               |
| Strategy         | Impact (high risk)<br>Score (1)     | No impact (high risk)<br>Score (1)                           | Optimize returns for risk: Score<br>[1] (1/1) |
| Example          | Low innovation (Maturity<br>stage)  | Innovation through<br>collaboration with other<br>industries |                                               |
| <b>Moderate</b>  |                                     |                                                              |                                               |
| Strategy         | No impact (low risk)<br>Score (0.5) | No impact (high risk)<br>Score (1)                           | Conservative: Score [0.5]<br>(0.5 / 1)        |
| <b>High Risk</b> |                                     |                                                              |                                               |
| Strategy         | Impact (high risk)<br>Score (1)     | Impact (low risk)<br>Score (0.5)                             | Speculative: Score [2]<br>(1/0.5)             |

# External Factors

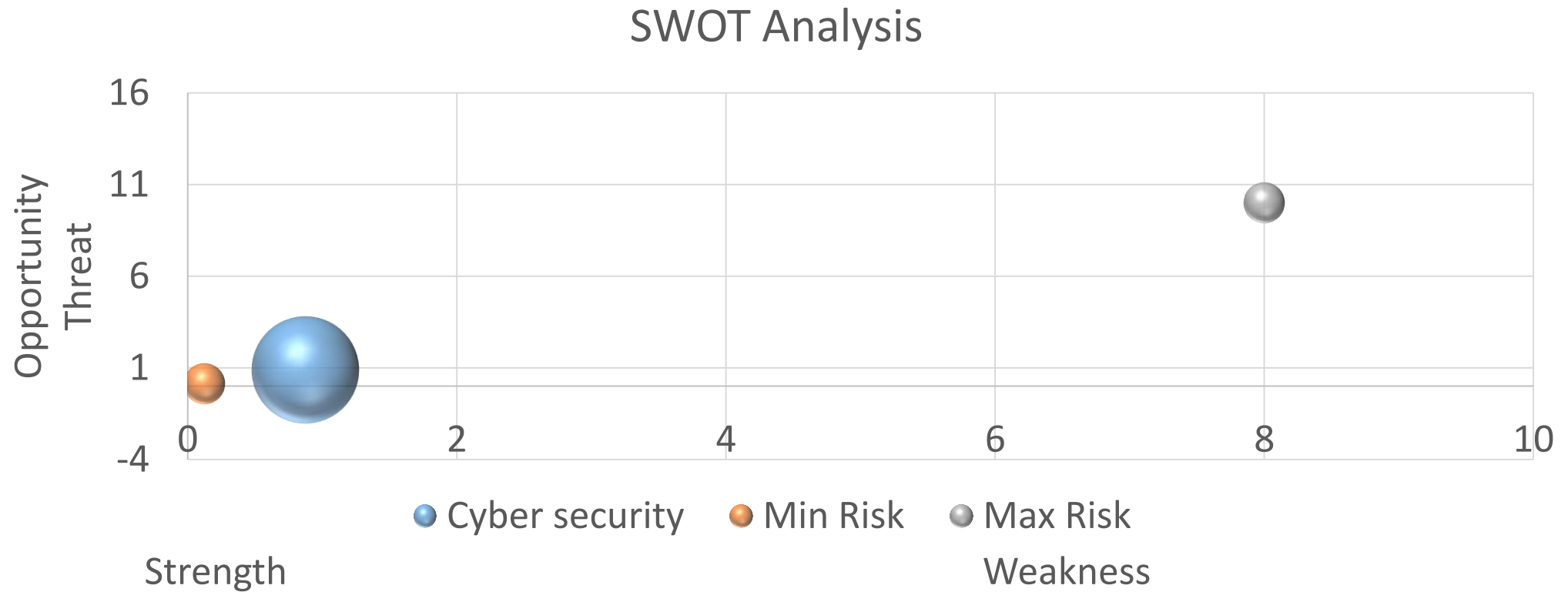
|               | Opportunity                        | Threat                              | Overall Score<br>(Threat/Opportunity) |
|---------------|------------------------------------|-------------------------------------|---------------------------------------|
| Low Risk      | Impact (low risk)<br>Score (0.5)   | No impact (low risk)<br>Score (0.5) | Score [1]<br>(0.5 / 0.5)              |
| Moderate Risk | Impact (low risk)<br>Score (0.5)   | Impact (high risk)<br>Score (1)     | Score 2]<br>(1 / 0.5)                 |
|               | No impact (high risk)<br>Score (1) | No impact (low risk)<br>Score (0.5) | Score [5]<br>(0.5 / 1)                |
| High Risk     | No impact (high risk)<br>Score (1) | Impact (high risk)<br>Score (1)     | Score [1]<br>(1 / 1)                  |

# SWOT: Impact (Risk) Evaluation (Cyber Security)

|   |                                     | Strength | Weakness | Opportunity | Threat |                    |
|---|-------------------------------------|----------|----------|-------------|--------|--------------------|
| 1 | Manufacturing cost coeff. Variation | 1        | 1.00     | 1           | 0.5    | BCI                |
| 2 | Total asset coeff. Variation        | 1        | 0.5      | 1           | 0.5    | Tax rate change    |
| 3 | Net profit coeff. Variation         | 0.5      | 0.5      | 1           | 0.5    | CCI                |
| 4 | sales coeff. Variation              | 1        | 0.5      | 0.5         | 1      | IT Industry return |
| 5 | Debt equity ratio covariance        | 0.5      | 1        | 0.5         | 1      | GDP growth         |

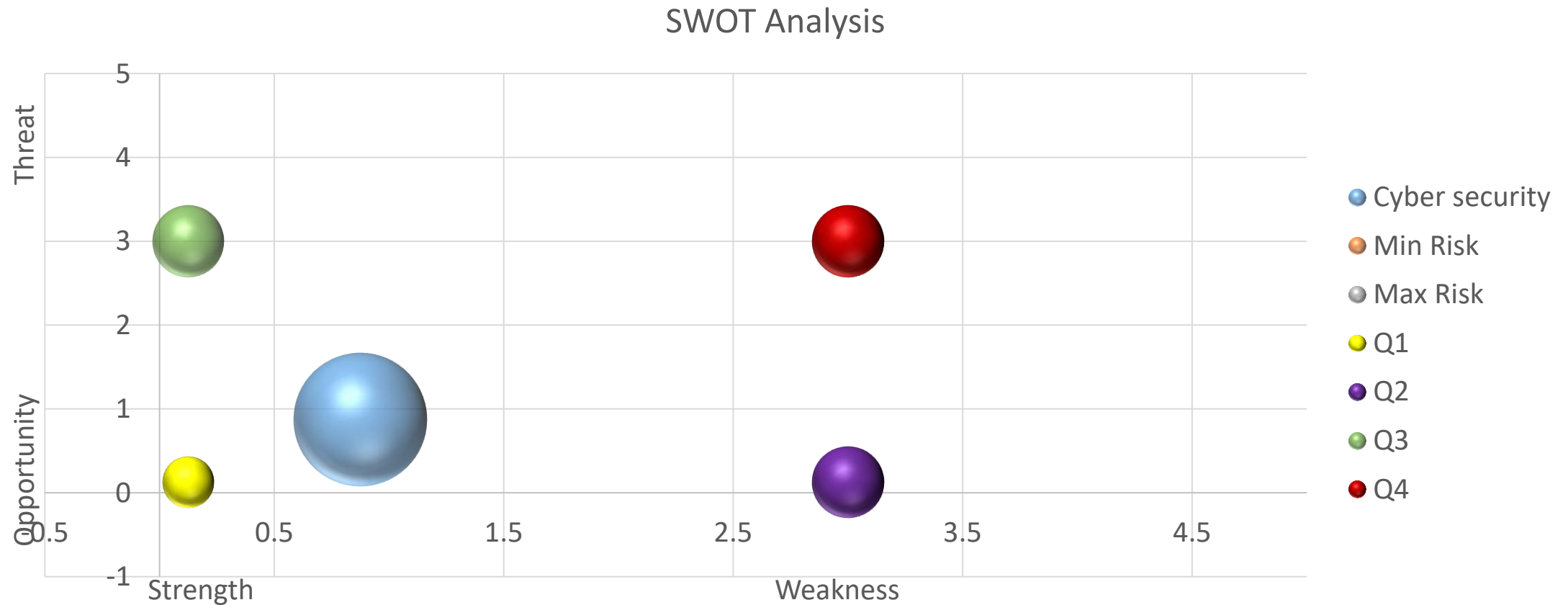
The key indicator for strength and opportunity (Impact) “I” is given a score of 1 and (Negative impact) “N” is given a score of 0.5. For weakness and threat; “I” is given 1 and “N” is given 0.5.

# SWOT Analysis (Cyber Security)





# SWOT Analysis (Cyber Security) Quadrants



# Cross Sectional Analysis

## **Factors**

- Finance
- Production
- Organization & Administration
- Marketing
- Technology
- Employee Skills
- Industry Risk

## **Representation**

- Good indicator of any industry
- Production intensive industry
- Automated industry
- Bulk commodity
- Standardized production process
- Unique representing potential
- Volatile pattern

# Cross Sectional Analysis Variables

| Factors               | Variable 1                          | Variable 2                             | Variable 3                             | Variable 4                |
|-----------------------|-------------------------------------|----------------------------------------|----------------------------------------|---------------------------|
| Product efficiency    | Raw material cost / total cost      | Production growth rate                 | Manufacturing cost / total cost        | Operating margin          |
| Market efficiency     | Advertisement cost / total sales    | HHI                                    | Sales growth                           | EPS growth rate           |
| Technical competence  | R&D/total sales                     | Cash flows from investment growth rate | Operating Cash flows/total assets      |                           |
| Financial performance | ROCE                                | Net profit /sales                      | Debt/Equity                            | DSCR                      |
| Employee skills       | Employee cost growth rate           | Managerial fees / total cost           | Employee cost / sales                  |                           |
| Administrative edge   | Reserves growth rate                | Foreign sales growth rate              | Operating cash flows/ operating profit | Dividend payout           |
| Industry risk         | Business risk (ROCE coef. variance) | Investment risk (ROA coef. variance)   | Total aseets coef. Variance            | Net profit coef. Variance |

# Product & Market Efficiency

| Weights | Factors                         | Cyber Security | Renewable Energy | Index Cyber Security | Index Renewable Energy |
|---------|---------------------------------|----------------|------------------|----------------------|------------------------|
| 0.25    | Raw material cost / total cost  | 0.000105       | -0.19443         | 0.131285             | -0.60906               |
| 0.25    | Production cost growth rate     | -0.0312        | 0.3466           | -0.4203              | 0.2964                 |
| 0.25    | Manufacturing cost / total cost | 0.1183         | 0.619742         | 0.29082              | 0.294298               |
| 0.25    | Operating margin                | 0.1870         | 0.214115         | 0.510459             | 0.817401               |

| Weights | Factors                          | Cyber Security | Renewable Energy | Index Cyber Security | Index Renewable Energy |
|---------|----------------------------------|----------------|------------------|----------------------|------------------------|
| 0.25    | Advertisement cost / total sales | 0.1657         | 0.006372         | 0.33818              | 0.263819               |
| 0.25    | HHI                              | 0.204711       | 0.203334         | 0.620688             | 0.27003                |
| 0.25    | Sales growth                     | 0.347566       | 0.327909         | 0.403256             | 0.255282               |
| 0.25    | EPS growth rate                  | 0.023017       | 0.46041          | 0.024675             | -0.08792               |

# Technical Competence & Employee Skill

| Weights | Factors                                | Cyber Security | Renewable Energy | Index Cyber Security | Index Renewable Energy |
|---------|----------------------------------------|----------------|------------------|----------------------|------------------------|
| 0.5     | R&D/total sales                        | 0.014467       | 2.156821         | 0.238711             | 0.192223               |
| 0.25    | Cash flows from investment growth rate | 1.502975       | 1.309701         | 0.128826             | 0.130076               |
| 0.25    | Operating Cash flows/total assets      | 0.66292        | 0.53333          | 0.63947              | 0.44213                |
| Weights | Factors                                | Cyber Security | Renewable Energy | Index Cyber Security | Index Renewable Energy |
| 0.5     | Employee cost growth rate              | -0.0943        | 0.087358         | -1.61709             | 0.30715                |
| 0.5     | Employee cost / total cost             | 0.4096         | 0.048975         | 0.65365              | 0.172197               |

# Financial & Administrative Edge

| Weights | Factors                                   | Cyber Security | Renewable Energy | Index Cyber Security | Index Renewable Energy |
|---------|-------------------------------------------|----------------|------------------|----------------------|------------------------|
| 0.35    | ROCE                                      | 0.224635       | 0.311529         | 0.581143             | 0.654095               |
| 0.35    | Net profit /sales                         | 1173.866       | 241.0557         | 0.382452             | 0.592249               |
| 0.15    | Debt/Equity                               | -0.02597       | -4.11989         | -0.03992             | -0.59846               |
| 0.15    | Interest coverage ratio                   | 38.36607       | 7.118724         | 0.424963             | 0.18914                |
| Weights | Factors                                   | Cyber Security | Renewable Energy | Index Cyber Security | Index Renewable Energy |
| 0.4     | Reserves growth rate                      | -0.14164       | 0.292536         | -0.05495             | 0.209044               |
| 0.5     | Operating cash flows/<br>operating profit | 0.6629         | 0.533333         | 0.639469             | 0.442129               |
| 0.1     | Dividend payout                           | 0.1671         | 0                | 0.230991             | 0                      |

# Industry Risk

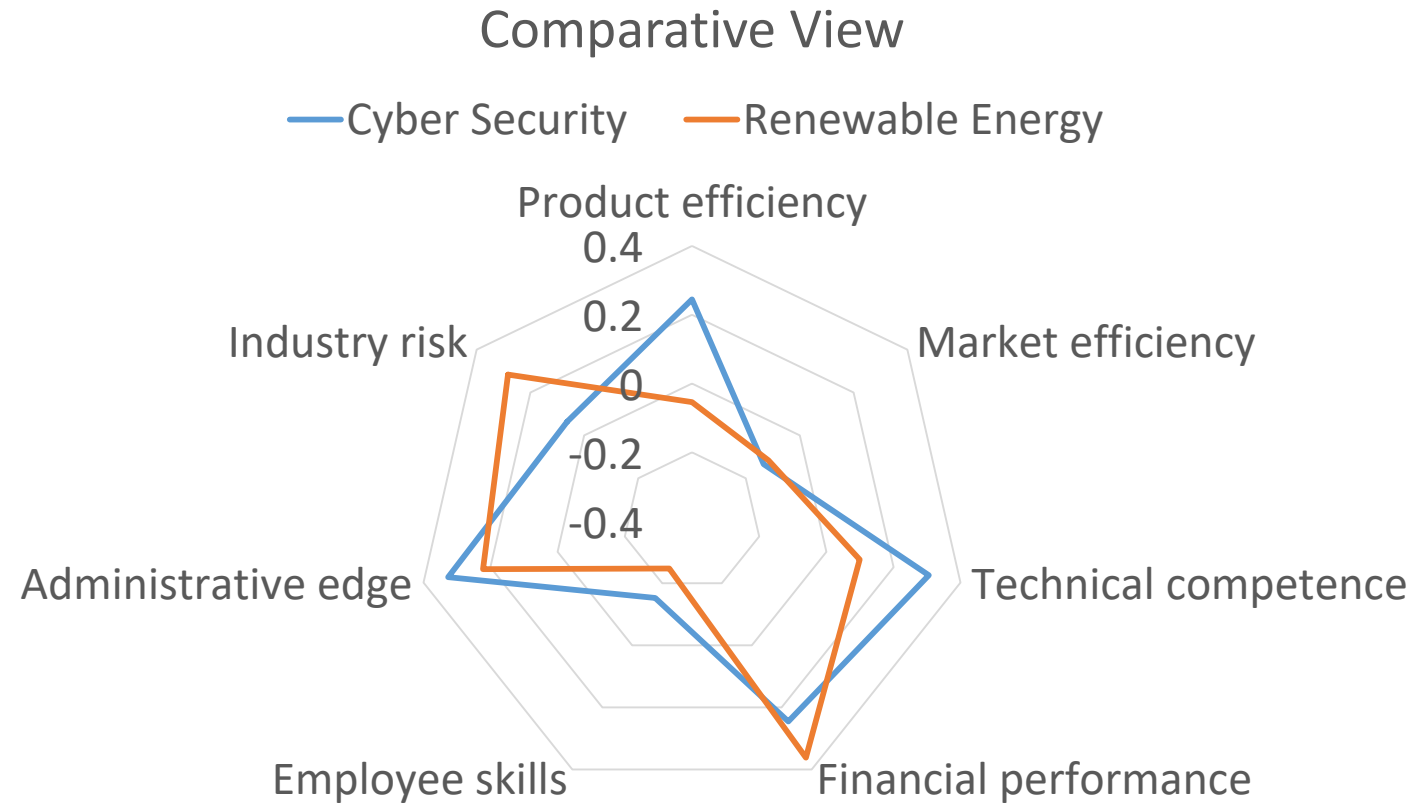
| Weights | Factors                              | Cyber Security | Renewable Energy | Index Cyber Security | Index Renewable Energy |
|---------|--------------------------------------|----------------|------------------|----------------------|------------------------|
| 0.25    | Business risk (ROCE coef. variance)  | -4.24784       | 1.231217         | -3.93412             | 0.674333               |
| 0.25    | Investment risk (ROA coef. variance) | 0.022131       | 1.242387         | 0.040711             | 0.169053               |
| 0.25    | Total aseets coef. Variance          | 0.339958       | 0.053832         | 0.353804             | 0.372792               |
| 0.25    | Net profit coef. Variance            | 0.062931       | 1.097768         | 0.065494             | 0.173653               |

# Cross Sectional Scores

| Factors               | Cyber Security | Renewable Energy |
|-----------------------|----------------|------------------|
| Product efficiency    | 0.244743       | -0.05419         |
| Market efficiency     | -0.13344       | -0.11586         |
| Technical competence  | 0.305432       | 0.099142         |
| Financial performance | 0.143665       | 0.25707          |
| Employee skills       | -0.15294       | -0.248           |
| Administrative edge   | 0.326706       | 0.222272         |
| Industry risk         | 0.06365        | 0.283491         |



# Cross Section View of Industries



# Investment Opportunity

## **Cyber Security Industry**

- High Risk; Better Financial performance
- High Production efficiency (high operating margin with low cost)
- High Technical competence
- Negative Market efficiency
- Suggestion: Long term investment

## **Renewable Energy Industry**

- Low Risk; Good Financial Performance
- Low operating margin compared to costs
- Average Technical competence
- Negative Market Efficiency
- Suggestion: Long term investment